

International Institute of Information Technology, Hyderabad
(Deemed to be University)

CS7.404: Digital Image Processing
IIT Hyderabad Question Paper
Mid Semester Examination

Max. Time: 1.5 Hr

Max. Marks: 20

Instructions

1. Answer all questions.
2. Make appropriate assumptions.
3. Place your Permanent / Temporary Student ID card on the desk during the examination for verification by the Invigilator.
4. Reading material such as books (unless open book exam) are not allowed inside the examination hall.
5. Borrowing writing material or calculators from other students in the examination hall is prohibited.
6. If any student is found indulging in malpractice or copying in the examination hall, the student will be given 'F' grade for the course and may be debarred from writing other examinations.

Additional sheet for rough work is allowed: Yes

1. **Operators:** Assume we have an image given by $f(x, y)$. Image multiplication can be defined as an operator which squares the intensity at each pixel location. ($5 + 2 = 7$)
 - a. Prove or disprove — Image multiplication is a linear operator.
 - b. If we have two images $f_1(x, y)$ and $f_2(x, y)$, both having the same resolution, multiplication would mean element-wise multiplication of pixel intensities, does your answer change?
2. **Bit-planes and Histograms:** We have bit plane slicing in the spatial domain, ($3 + 3 + 3 = 9$)
 - a. Can there be an equivalent operation in the Fourier/frequency domain? If yes, define the operation in a rigorous manner, if no, then provide reasons for non existence.
 - b. How would the histogram of an image change if we set lower-order bit planes to zero?
 - c. If we set higher-order bit planes to zero?
3. **Frequency domain transformations:** The second image was obtained by: (a) multiplying the first image by $(-1)^{x+y}$; (b) computing the DFT; (c) taking the complex conjugate of the transform; (d) computing the inverse DFT; and (e) multiplying the real part of the result by $(-1)^{x+y}$. Explain (mathematically) why the image on the right appears as it does. (4)

CLASSMATE

Image 1



Image 2

